POLYNOMIAL MINIMAL GRAPHS OF DEGREE AT MOST FOUR ARE AFFINE

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Abstract. We give a self-contained proof that every real scalar polynomial of total degree at most four whose graph is a minimal hypersurface is affine, in every dimension. The quartic step is a sign obstruction for homogeneous solutions of the polynomial 1-Laplacian. A coefficient identity at a spherical critical point expresses the Laplacian as a cubic polynomial in a tangential symmetric matrix. Comparing a positive maximum and a negative minimum forces opposite signs on the same mixed coefficient. A flux argument shows that the leading homogeneous part of a nonlinear polynomial minimal graph must change sign, giving the contradiction. Elementary proofs of the known quadratic and cubic exclusions complete the argument. We describe the relation to earlier p-Laplacian coefficient methods and provide a single-file Lean formalization of the full degree-at-most-four statement, checked against the corresponding Conjectures.io target.

1. Introduction and main result

For a twice differentiable function $u\colon \mathbb{R}^n \to \mathbb{R}$, set

$$\mathcal{L}u = |\nabla u|^2 \Delta u - \langle \nabla u, (\operatorname{Hess} u)\nabla u\rangle, \qquad \mathcal{M}u = \Delta u + \mathcal{L}u. \tag{1}$$

All norms and inner products are Euclidean. The graph of u is a minimal hypersurface precisely when

$$\mathcal{M}u = (1 + |\nabla u|^2)\Delta u - \sum_{i,j=1}^{n} u_i u_j u_{ij} = 0. \tag{2}$$

Equivalently,

$$\operatorname{div}\left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}}\right) = 0. \tag{3}$$

Here and throughout, the graph has codimension one; the analogous question for vector-valued polynomial maps is different.

Theorem 1.1. *Let $n \geq 1$ and let $u \in \mathbb{R}[x_1, \ldots, x_n]$ have total degree at most four. If $\mathcal{M}u = 0$ identically, then u is affine.*

The assertion is also true for $n = 0$, when every polynomial is constant. Including this case gives the quantification used in the formal statement in Section 6. Conversely, every affine polynomial satisfies (2).

The broader question of whether a nonlinear polynomial can solve the minimal surface equation goes back to Simon; see the discussion and reference in Guo [1]. Guo proves a structure theorem for polynomial minimal graphs, excludes degrees two and three, and obtains further constraints on the leading homogeneous part and the tangent cone at infinity. His stated general degree bound is $\deg u \geq 4$ for a nonlinear solution. In dimension eight, his factorization and tangent-cone results already imply that the degree is odd. Together with the classical Bernstein theorem in dimensions at most seven, this places the additional dimensional coverage of Theorem 1.1 at $n \geq 9$; see Remark 5.1.

The local identity used in our quartic argument is a specialization of a coefficient identity in Lewis–Vogel [2]. Their stated quartic classification excludes the parameter $p = 1$, but their proof admits a short

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endpoint adaptation, discussed in Section 5. The argument here uses two spherical extrema to obtain just the sign obstruction needed for the minimal-graph problem, without carrying out that classification. The homogeneous cubic classification used below is already explicit in Tkachev [3, Remark 3.2]; we reprove it to keep the proof self-contained. We make no claim of historical priority for the theorem or for the local coefficient identity.

The proof has three parts. First, the leading homogeneous part of any nonlinear polynomial solution satisfies $\mathcal{L}p = 0$ and must change sign. Second, a homogeneous quartic satisfying $\mathcal{L}p = 0$ cannot change sign. Finally, direct calculations exclude the two lower nonlinear degrees.

2. The leading term and a flux obstruction

Lemma 2.1. *Let u be a polynomial of degree $m \geq 2$ with leading homogeneous part p. If $\mathcal{M}u = 0$, then $\mathcal{L}p = 0$.*

Proof. The homogeneous component of degree $3m - 4$ in $\mathcal{L}u$ is $\mathcal{L}p$. Every term containing a lower-degree component of u has smaller degree. Since $\deg \Delta u \leq m - 2 < 3m - 4$, the component of degree $3m - 4$ in $\mathcal{M}u$ is exactly $\mathcal{L}p$. □

Lemma 2.2 (Flux obstruction)**.** *Under the hypotheses of Lemma 2.1, the polynomial p takes both positive and negative values.*

Proof. It suffices to rule out $p \geq 0$, since $-u$ also satisfies (2). Suppose, therefore, that $p \geq 0$. For $R > 0$, define

$$V_R(x) = \frac{\nabla u(Rx)}{\sqrt{1 + |\nabla u(Rx)|^2}}.$$

Equation (3) gives $\operatorname{div}_x V_R = 0$ and $|V_R| \leq 1$. Choose the smooth compactly supported function

$$\varphi(x) = \begin{cases} -\exp\bigl(-1/(1 - |x|^2)\bigr), & |x| < 1, \\ 0, & |x| \geq 1. \end{cases}$$

It satisfies $\nabla\varphi(x) = w(x)x$, where $w(x) = 2\exp(-1/(1 - |x|^2))/(1 - |x|^2)^2 > 0$ for $|x| < 1$ and $w = 0$ outside the unit ball. Integration by parts yields

$$\int_{\mathbb{R}^n} \langle \nabla\varphi, V_R \rangle \, dx = 0. \tag{4}$$

The zero set of a nonzero real polynomial has Lebesgue measure zero. For completeness, this follows by induction on the number of variables: write the polynomial as a polynomial in the last variable, discard the measure-zero common zero set of its coefficient polynomials, and apply the one-variable root bound on each remaining fiber, followed by Fubini's theorem. Thus $p(x) > 0$ for almost every x. Euler's identity gives

$$\langle x, \nabla p(x) \rangle = m p(x) > 0 \tag{5}$$

at each such point, so $\nabla p(x) \neq 0$ there. Since $m \geq 2$,

$$R^{1-m} \nabla u(Rx) \longrightarrow \nabla p(x), \qquad V_R(x) \longrightarrow \frac{\nabla p(x)}{|\nabla p(x)|}$$

whenever $p(x) > 0$. The integrands in (4) are bounded in absolute value by the integrable function $|\nabla\varphi|$. Dominated convergence therefore gives

$$0 = \int_{|x|<1} w(x) \frac{m p(x)}{|\nabla p(x)|} \, dx > 0,$$

where the quotient may be defined arbitrarily on its measure-zero exceptional set. The strict inequality holds because the integrand is positive almost everywhere in the unit ball. This is a contradiction. □

3. The quartic sign obstruction

The next identity is the algebraic input. If S is a symmetric operator and $a \neq 0$, the notation $F_a(S)$ means the matrix polynomial

$$F_a(S) = 2S + \frac{S^2}{a} + \frac{S^3}{2a^2}. \tag{6}$$

Lemma 3.1 (Identity at a spherical critical point)**.** *Let p be a homogeneous real quartic on $\mathbb{R}^n$ satisfying $\mathcal{L}p = 0$. Let e be a unit vector that is a critical point of the restriction of p to the unit sphere, and suppose $p(e) = a \neq 0$. In coordinates $x = te + y$ with $y \in e^\perp$, write*

$$p(te + y) = at^4 + t^2A(y) + tB(y) + C(y), \tag{7}$$

where A, B, C are homogeneous of degrees two, three, and four, respectively. There is a symmetric operator S on $e^\perp$ such that $A(y) = \langle y, Sy\rangle$, and

$$\Delta p(te + y) = 12at^2 + \langle y, F_a(S)y\rangle. \tag{8}$$

Moreover, $\Delta_y A = \Delta_y B = 0$.

Proof. Criticality and Euler's identity imply $\nabla p(e) = 4ae$. Consequently the t^3 term, which would be linear in y, vanishes, giving (7). Put $\alpha = \nabla_y A$ and $W = |\nabla p|^2$. In the following coefficient calculation, derivatives of A, B, C are taken only in the y variables. We have

$$\begin{aligned}\Delta p &= (12a + \Delta A)t^2 + (\Delta B)t + 2A + \Delta C,\\ W &= 16a^2t^6 + (16aA + |\alpha|^2)t^4 + (8aB + 2\langle\alpha, \nabla B\rangle)t^3 + \text{terms of } t\text{-degree at most two.}\end{aligned}$$

Use the identity

$$\mathcal{L}p = W\Delta p - \tfrac{1}{2}\langle\nabla p, \nabla W\rangle.$$

The coefficients of t^8 and t^7 are, respectively, $16a^2\Delta A$ and $16a^2\Delta B$. They vanish, so $\Delta A = \Delta B = 0$. After these identities are imposed, the coefficients of t^6 in the two terms are

$$\begin{aligned}[t^6](W\Delta p) &= 224a^2A + 16a^2\Delta C + 12a|\alpha|^2,\\ [t^6]\left(\tfrac{1}{2}\langle\nabla p, \nabla W\rangle\right) &= 224a^2A + 16a|\alpha|^2 + \langle\alpha, (\operatorname{Hess} A)\alpha\rangle.\end{aligned}$$

Here $[t^j]$ denotes the coefficient of t^j, with polynomial coefficients in y. Subtracting gives

$$16a^2\Delta C - 4a|\nabla A|^2 - \langle\nabla A, (\operatorname{Hess} A)\nabla A\rangle = 0. \tag{9}$$

Since $\nabla A = 2Sy$ and $\operatorname{Hess} A = 2S$, this is

$$\Delta C = \left\langle y, \left(\frac{S^2}{a} + \frac{S^3}{2a^2}\right)y\right\rangle.$$

Substituting into $\Delta p = 12at^2 + 2A + \Delta C$ proves (8). □

Lemma 3.2. *For every real $a \neq 0$, the function $F_a(s) = 2s + s^2/a + s^3/(2a^2)$ is strictly increasing on $\mathbb{R}$, and $\operatorname{sgn} F_a(s) = \operatorname{sgn} s$.*

Proof. The identities

$$F_a(s) = \frac{s}{2}\left[\left(\frac{s}{a} + 1\right)^2 + 3\right], \qquad F_a'(s) = \frac{3}{2}\left(\frac{s}{a} + \frac{2}{3}\right)^2 + \frac{4}{3}$$

give both assertions, for either sign of a. □

Proposition 3.3 (Quartic sign obstruction)**.** *If p is a homogeneous real quartic satisfying $\mathcal{L}p = 0$, then p is everywhere nonnegative or everywhere nonpositive.*

Proof. Suppose that p takes both signs. By compactness of the unit sphere and homogeneity, there are unit vectors e, f at which its maximum and minimum are attained, with

$$p(e) = a > 0, \qquad p(f) = b < 0.$$

Let $Q = \frac{1}{2}\operatorname{Hess}(\Delta p)$, the constant symmetric matrix representing the homogeneous quadratic polynomial Δp: $\Delta p(x) = \langle x, Qx\rangle$. Lemma 3.1 at e and at f gives

$$Qe = 12ae, \qquad Qf = 12bf.$$

The eigenvalues are distinct. Symmetry of Q therefore implies $e \perp f$.

Use the normal form (7) at e, with tangential operator S. On $e^{\perp}$, equation (8) identifies Q with $F_a(S)$. Hence

$$F_a(S)f = 12bf. \tag{10}$$

Diagonalize S orthogonally. Each nonzero component of f in an eigenspace of S has an eigenvalue λ satisfying $F_a(\lambda) = 12b$. Strict monotonicity of F_a shows that all these eigenvalues coincide. Thus $Sf = \lambda f$, and the sign assertion in Lemma 3.2 gives $\lambda < 0$. The coefficient of t^2s^2 in $p(te + sf)$ is therefore

$$A(f) = \langle f, Sf\rangle = \lambda < 0. \tag{11}$$

Interchanging e and f, let T be the tangential operator in the normal form at f. The same reasoning gives $Te = \mu e$ with $F_b(\mu) = 12a$, hence $\mu > 0$. The coefficient of s^2t^2 in $p(sf + te)$ is $\langle e, Te\rangle = \mu > 0$. These are the same coefficient in the same bivariate polynomial, contradicting (11). □

Corollary 3.4. *No polynomial of degree exactly four satisfies $\mathcal{M}u = 0$.*

Proof. Its leading homogeneous part would satisfy $\mathcal{L}p = 0$ by Lemma 2.1, would take both signs by Lemma 2.2, and would have one sign by Proposition 3.3. □

4. The lower degrees

The quadratic and cubic exclusions are known [1]. We include direct proofs using the same polynomial operator.

Proposition 4.1. *No polynomial of degree exactly two satisfies $\mathcal{M}u = 0$.*

Proof. Write $u(x) = \frac{1}{2}\langle x, Hx\rangle + \langle c, x\rangle + d$, where H is real symmetric and nonzero, and put $s = \operatorname{tr} H$. The quadratic part of $\mathcal{M}u = 0$ is

$$H^2(sI - H) = 0.$$

Every nonzero eigenvalue of H is therefore equal to s. If $s = 0$, all eigenvalues vanish, which contradicts $H \neq 0$. Otherwise, taking the trace gives $s = (\operatorname{rank} H)s$, so $\operatorname{rank} H = 1$. Decompose c orthogonally as a sum of vectors in $\operatorname{im} H$ and $\ker H$, and choose x so that $Hx + c = c_{\perp} \in \ker H$. Evaluating the equation at this x gives

$$0 = (1 + |c_{\perp}|^2)s - \langle c_{\perp}, Hc_{\perp}\rangle = (1 + |c_{\perp}|^2)s,$$

again a contradiction. □

Lemma 4.2 (Homogeneous cubic classification)**.** *A nonzero homogeneous real cubic p satisfying $\mathcal{L}p = 0$ is a nonzero constant times the cube of a real linear form.*

Proof. Since p is odd, it has a positive maximum a on the unit sphere, attained at some e. Criticality gives, in orthogonal coordinates $x = te + y$,

$$p(te + y) = at^3 + tA(y) + B(y),$$

with A quadratic and B cubic. The coefficients of t^5 and t^4 in $\mathcal{L}p$ are $9a^2\Delta A$ and $9a^2\Delta B$, respectively. Consequently both Laplacians vanish. The coefficient of t^3 then reduces to $-\langle \nabla A, (\operatorname{Hess} A)\nabla A\rangle$. Writing $A(y) = \langle y, Sy\rangle$ with S symmetric gives $\langle y, S^3y\rangle = 0$ for every y. Thus $S^3 = 0$, and the spectral theorem implies $S = 0$.

Now $p = at^3 + B(y)$ and $\Delta B = 0$. Direct differentiation gives

$$\mathcal{L}p = 6at|\nabla B|^2 - \langle \nabla B, (\operatorname{Hess} B)\nabla B\rangle.$$

The coefficient of t forces $\nabla B = 0$; homogeneity then gives $B = 0$. Thus $p = at^3$ in these coordinates. □

Proposition 4.3. *No polynomial of degree exactly three satisfies $\mathcal{M}u = 0$.*

Proof. By Lemmas 2.1 and 4.2, an orthogonal change of coordinates makes the leading part at^3, where $a \neq 0$. The full polynomial can therefore be written as

$$u(t, y) = at^3 + bt^2 + t(\beta \cdot y + c) + \langle y, Sy\rangle + d \cdot y + e_0,$$

where S is symmetric. Orthogonal changes of coordinates preserve (2). Put $\ell = \beta \cdot y + c$ and $w = 2Sy + d$. Then $u_t = 3at^2 + 2bt + \ell$, $\nabla_y u = t\beta + w$, and the coefficient of t^4 in $\mathcal{M}u$ is $18a^2 \operatorname{tr} S$. Hence $\operatorname{tr} S = 0$. With this trace zero, the equation becomes

$$\begin{aligned} 0 = {} & (6at + 2b)(1 + |t\beta + w|^2) \\ & - 2(3at^2 + 2bt + \ell)\beta \cdot (t\beta + w) - 2\langle t\beta + w, S(t\beta + w)\rangle. \end{aligned} \tag{12}$$

Its coefficient of t^2 is

$$6a\beta \cdot w - 2b|\beta|^2 - 2\langle \beta, S\beta\rangle.$$

The terms linear in y imply $S\beta = 0$. The coefficient of t in (12) consequently simplifies to

$$6a(1 + |2Sy + d|^2) - 2|\beta|^2(\beta \cdot y + c) = 0. \tag{13}$$

Its quadratic part is $24a\langle y, S^2 y\rangle$, so $S = 0$. Its linear part now gives $|\beta|^2\beta = 0$, hence $\beta = 0$. Finally its constant part gives $6a(1 + |d|^2) = 0$, which is impossible. □

Proof of Theorem 1.1. Corollary 3.4 and Propositions 4.1 and 4.3 exclude degrees four, two, and three. A polynomial of degree at most one is affine. □

5. Relation to previous results

The polynomial operator associated with the p-Laplacian is

$$|\nabla v|^2 \Delta v + (p - 2)\langle \nabla v, (\operatorname{Hess} v)\nabla v\rangle.$$

At $p = 1$ this is $\mathcal{L}v$. Lewis–Vogel [2, Theorem 1.2] classify homogeneous quartic solutions for parameters excluding 1 and 2. Their equation (2.13), specialized to degree four and $p = 1$, is the local identity in Lemma 3.1: with their normalization $a = 1/12$ and $S = \operatorname{diag}(a_i/2)$,

$$F_{1/12}(a_i/2) = a_i + 3a_i^2 + 9a_i^3.$$

Their proof also indicates an endpoint classification after a short adaptation. Their equations (3.4)–(3.9) eliminate the cubic transverse term. At $p = 1$, retain the all-a_i-zero case discarded just before (3.14); equation (3.11) then eliminates the quartic transverse term. Otherwise, the argument through (3.20) makes the smallest nonzero a_i equal to $1/3$, contradicting $\sum a_i = 0$ from (2.11). This is an observation about their proof, rather than a statement of their theorem. The proof of Proposition 3.3 does not use this elimination: it compares two extrema and proves only sign-definiteness.

Remark 5.1. Guo [1, Theorems 1.3 and 1.7] writes the leading part of a nonlinear polynomial solution as $p^k Q$, where k is odd and Q has even degree. In dimension eight, $\deg p = 3$, so the total degree is odd. His exclusion of cubic solutions therefore already rules out degrees at most four in that dimension. For $n \leq 7$, the classical Bernstein theorem rules out every nonlinear entire minimal graph; see Guo's introduction for the history. The proof above applies uniformly in all dimensions and does not use those geometric results.

6. Lean formalization and verification

The accompanying file `QuarticMinimalGraphs.lean` formalizes the degree-at-most-four conclusion over real multivariate polynomials, with variables indexed by `Fin n`. The formal target is `Math30Catalog.source28` in the Conjectures.io Math 30 catalog [4]. Its mathematical content is

$$\forall n \in \mathbb{N}\ \forall P \in \mathbb{R}[X_i \mid i \in \mathrm{Fin}(n)], \quad \deg P \le 4 \ \wedge\ \mathcal{M}P = 0 \implies \deg P \le 1,$$

where degree is the library's natural-number-valued total degree, including the zero polynomial. The operator is defined by formal partial derivatives and finite sums, with the same expression as (2).

The submission body ends with the declaration

```
theorem target : fcTypeOfName% "Math30Catalog.source28" := by
  intro n P hdegree hminimal
  exact QuarticMinimal.affine_of_degree_le_four_minimal
    P hdegree hminimal
```

The trusted challenge header supplies

```
import FormalConjectures.ResearchTargets.Math30
import TaskSupport

namespace Bounty
```

and its footer closes the namespace. Thus the delivered Lean file is the single solution body expected by that challenge, rather than a standalone project with its own imports. The type macro reads the target's type; the proof does not use the catalog's admitted target as a premise.

The formal development includes the homogeneous leading-term reduction, orthogonal normalization, coefficient comparisons, the quartic sign obstruction, the quadratic and cubic exclusions, and the analytic flux argument. In the latter, it uses a compactly supported radial test function, integration by parts, polynomial asymptotics, and dominated convergence, as in Lemma 2.2. Some intermediate linear algebra is implemented by positive quadratic identities rather than explicit spectral diagonalization. The formalization proves the same result without requiring the source proof's exact sequence of intermediate lemmas.

The official local verifier returned `VERIFIED` for the exact submission bytes. It checked equality with the required statement, submission-policy compliance, and acceptance by the Lean kernel. The proof uses no `sorry`, `admit`, or additional axioms; its reported axiom dependencies are the standard `propext`, `Classical.choice`, and `Quot.sound`. The checked environment used Lean 4.35.0-rc2. The verification report and exact dependency revisions are included with the companion artifact. Verification was run in the checker's native macOS development mode; production sandbox certification was not performed. No claim of external submission acceptance or of historical novelty follows from this check.

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The proof development, literature search, coefficient calculations, expository drafting, and Lean formalization were assisted by OpenAI Codex. This assistance is disclosed separately from the author list. The Lean kernel check certifies the formal statement in the recorded environment; it does not certify bibliographic priority or the correspondence of every sentence of the exposition with the formal artifact. The formalization contains attributed adaptations of auxiliary Mathlib proofs and retains their Apache 2.0 licensing notices. We acknowledge the earlier results of Guo, Lewis–Vogel, and Tkachev that provide the context and coefficient methods discussed above.

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